

Basavaraja M

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Summary

Motivated Electronics and Communication Engineering student with good knowledge of Python, Flask, OOPS, and MySQL, eager to build a career in software development. Passionate about learning new technologies, improving problem-solving skills, and contributing to innovative teams.

Education

RNS Institute of Technology	2022 – 2026
B.E. in Electronics and Communication Engineering	CGPA: 8.53
Akshara PU College, Shimoga	2020 – 2022
12th, Karnataka PUE Board	92%
Govt Junior College, Shiralakoppa	2018 – 2020
10th, KSEEB	90.88%

Skills

Programming: Python, C, Embedded C, OOPS

Frameworks: Flask

Software: Data Structures and Algorithms, Operating Systems, Computer Networks, Linux, HTML, CSS, JavaScript, MySQL, Basics of Gen AI

Tools: Verilog, MATLAB, Xilinx Vivado

Projects

Library Management System Using Python Flask and MySQL

- Developed a web-based Library Management System using Python (Flask) and MySQL to manage books and issuing operations.
- Implemented CRUD functionality with database connectivity and real-time quantity validation.
- Followed SDLC principles and tested modules individually, strengthening Python programming and SQL skills.

Implementation of L1 Cache on VexRISC-V Processor

Apr 2025 – Aug 2025

- Implemented L1 Instruction and Data Cache on the VexRISC-V processor using SpinalHDL plugins.
- Configured cache parameters (size, line length, associativity) to improve hit rate and reduce memory latency.
- Enhanced overall processor performance by optimizing instruction fetch and data access in an FPGA-based system.

Vehicle Detection, Counting & Tracking using YOLOv3

Jan 2025 – Jun 2025

- Developed a vehicle detection, counting, and tracking system using YOLOv3 for real-time traffic analysis.
- Implemented object detection and tracking algorithms to identify vehicles and maintain unique IDs across frames.
- Improved accuracy and performance through model tuning and optimized video processing pipeline.

Certifications

- Programming Fundamentals using Python – Infosys Springboard
- MATLAB – Advanced
- Generative AI: Introduction and Applications - Coursera
- VLSI for Beginners – NIT Calicut
- Pre-Silicon RTL Verification (IEEE 1800.2-2020) – KSSEM
- SQL (Basic) Certificate – HackerRank

Achievements & Activities

- Solved multiple Verilog problems on HDLBits, strengthening understanding of digital design and RTL concepts.
- Actively participated in college sports events including 5km running and cricket tournaments.